

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

Student Name: S Satya Jayavanth

Roll No.: A24126552116

Date: 29-08-2026

1. TITLE

Design Static Webpage using HTML Components

2. AIM

To design and develop a structured static webpage using HTML5 semantic elements and HTML Components.

3. OBJECTIVE

The objectives of this experiment are:

- ☐ To understand the basic structure of an HTML5 document.
 - ☐ To create a static webpage using HTML components.
 - ☐ To use semantic HTML elements such as <header>, <nav>, <main>, <section>, <article>, <aside> and <footer>.
 - ☐ To create navigation links within a webpage.
 - ☐ To create a user inquiry form using HTML form controls.
 - ☐ To organize webpage information using ordered and unordered lists.
-

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. A static webpage contains fixed content that is displayed directly by a web browser.

HTML5 provides semantic elements that clearly describe the purpose of different parts of a webpage. The <header> element is used for the introductory section, <nav> contains navigation links, <main> contains the primary content, <section> groups related content, <article> represents independent content, <aside> contains supplementary information, and <footer> represents the closing section of the webpage.

HTML also provides various components for collecting user input. Form controls such as text boxes, email fields, drop-down lists, text areas, submit buttons and reset buttons can be created using HTML. In this experiment, these components are used to design a complete static webpage without external CSS.

5. ALGORITHM / PROCEDURE

- ☐ Create a new HTML file and declare the HTML5 document type using <!DOCTYPE html>.
 - ☐ Create the <html>, <head> and <body> sections.
 - ☐ Add the webpage title inside the <title> element.
 - ☐ Create a header containing the webpage name and navigation links.
 - ☐ Create a home section containing a heading, description and navigation link.
 - ☐ Create the main content area using HTML structural components.
 - ☐ Add articles describing HTML structural and interaction elements.
 - ☐ Create a user inquiry form using HTML form controls.
 - ☐ Add text input, email input, select, textarea, submit and reset controls.
 - ☐ Create a sidebar using the <aside> element.
 - ☐ Add ordered and unordered lists to organize information.
 - ☐ Create a footer containing copyright information.
 - ☐ Save the HTML file and open it in a web browser.
 - ☐ Verify the webpage structure, navigation links and form controls.
 - ☐ Compare the actual output with the expected output.
-

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Pure Semantic HTML Static Webpage</title>
</head>
<body>

  <!-- Header Section -->
  <header>
    <table width="100%" border="0" cellspacing="0" cellpadding="10">
```

```

<tr>
  <td>
    <h1>StaticWeb</h1>
  </td>
  <td align="right">
    <nav>
      <a href="#home">Home</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
      <a href="#articles">Articles</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
      <a href="#about">About</a> &nbsp;&nbsp;&nbsp;|&nbsp;&nbsp;&nbsp;
      <a href="#contact">Contact Us</a>
    </nav>
  </td>
</tr>
</table>
</header>

<hr>

<!-- Hero / Showcase Section -->
<section id="home">
  <center>
    <h2>Semantic Structures Without CSS</h2>
    <p>This layout uses raw HTML components to build an organized document hierarchy
that browsers can cleanly parse out-of-the-box.</p>
    <p><a href="#articles"><b>Read Our Articles Below &darr;</b></a></p>
  </center>
</section>

<hr>

<!-- Main Content Layout using a Structural Table -->
<table width="100%" border="0" cellspacing="0" cellpadding="20">
  <tr valign="top">

    <!-- Left Side: Main Content Column -->
    <td width="70%">
      <main id="articles">

        <article>
          <h2>1. Core Structural Tags</h2>
          <p>Published: July 7, 2026</p>
          <p>When you omit cascading style sheets entirely, the semantic markup handles
all of the structural heavy lifting. Elements like <code>&lt;main&gt;</code>,

```

`<article>`, and `<section>` tell screen readers and web scrapers exactly how information flows.

Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

`</article>`

`<br><hr><br>`

`<article>`

`<h2>2. Native Interaction Elements</h2>`

`<p>Published: July 5, 2026</p>`

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

`<h3>Interactive Demonstration Forms</h3>`

`<form id="contact" action="#" method="post">`

`<fieldset>`

`<legend><b>User Inquiry Form</b></legend>`

`<p>`

`<label for="name">Full Name: </label>`

`<input type="text" id="name" name="username" required
placeholder="John Doe">`

`</p>`

`<p>`

`<label for="email">Email Address: </label>`

`<input type="email" id="email" name="useremail" required>`

`</p>`

`<p>`

`<label for="topic">Inquiry Topic: </label>`

`<select id="topic" name="usertopic">`

`<option value="general">General Feedback</option>`

`<option value="technical">Technical Architecture</option>`

`<option value="academics">Academic Curriculum</option>`

`</select>`

`</p>`

`<p>`

`<label for="message">Message Details:</label><br>`

`<textarea id="message" name="usermessage" rows="5" cols="40"
placeholder="Type your notes here..."></textarea>`

`</p>`

`<p>`

`<input type="submit" value="Submit Form Data">`

`<input type="reset" value="Clear Entries">`

```

        </p>
        </fieldset>
    </form>
</article>

</main>
</td>

<!-- Right Side: Sidebar Column -->
<td width="30%" bgcolor="#f4f4f4">
    <aside id="about">
        <h3>Document References</h3>
        <p>Static pages map data out directly without real-time database queries or
processor overhead.</p>

        <h4>Core Benefits:</h4>
        <ul>
            <li>Instant browser painting</li>
            <li>Low memory footprints</li>
            <li>High data compatibility</li>
        </ul>

        <h4>Required Components:</h4>
        <ol>
            <li>Document Type Definition</li>
            <li>Semantic Wrapper Blocks</li>
            <li>Data Input Control Interfaces</li>
        </ol>
    </aside>
</td>

</tr>
</table>

<hr>

<!-- Footer Section -->
<footer>
    <center>
        <p>&copy; 2026 Pure HTML Architecture. Assembled completely without style
configurations.</p>
    </center>
</footer>

```

```
</body>
</html>
```

7. CODE EXPLANATION

<!DOCTYPE html> – Declares that the document uses HTML5.

<html> – Defines the root element of the HTML document.

<head> – Contains metadata and the title of the webpage.

<body> – Contains the visible content of the webpage.

<header> – Defines the header section of the webpage.

<nav> – Contains navigation links such as Home, Articles, About and Contact Us.

<section> – Groups related content into a logical section.

<main> – Defines the main content of the webpage.

<article> – Represents an independent block of content.

<form> – Creates a form for collecting user information.

<input> – Creates different input controls such as text, email, submit and reset fields.

<select> – Creates a drop-down list containing different options.

<textarea> – Creates a multi-line text input area.

<fieldset> – Groups related form controls together.

<legend> – Provides a caption for the fieldset.

<aside> – Defines supplementary or sidebar information.

**** – Creates an unordered list.

**** – Creates an ordered list.

<footer> – Defines the footer section of the webpage.

8. TEST CASES AND EXECUTION DATA

I would not just write three random test cases. I would actually execute them and record the results.

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Open the HTML file in a web browser	Static webpage should be displayed correctly	Static webpage displayed correctly	Pass

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-02	Click the Home navigation link	Home section should be displayed	Home section displayed	Pass
TC-03	Click Articles and About links	Corresponding sections should be displayed	Corresponding sections displayed	Pass
TC-04	Enter valid name, email, topic and message in the form	Form should accept the entered information	Information accepted	Pass
TC-05	Click Clear Entries	All form fields should be cleared	Form fields cleared	Pass
TC-06	Leave required fields empty and submit	Browser should display required-field validation	Validation displayed	Pass

9. OUTPUT

Semantic Structures Without CSS

This layout uses raw HTML components to build an organized document hierarchy that browsers can cleanly parse out-of-the-box.

[Read Our Articles Below.](#)

1. Core Structural Tags

Published: July 7, 2026

When you omit cascading style sheets entirely, the semantic markup handles all of the structural heavy lifting. Elements like <main>, <article>, and <section> tell screen readers and web scrapers exactly how information flows.

Using raw browser defaults ensures excellent loading performance, perfect accessibility compliance, and absolute structural clarity.

2. Native Interaction Elements

Published: July 5, 2026

Browsers come equipped with excellent interactive elements by default. We can capture text inputs, provide multiple-choice configurations, and submit structured information to backend parsers using pure markup.

Interactive Demonstration Forms

User Inquiry Form

Full Name:

Email Address:

Inquiry Topic:

General Feedback

Message Details:

Type your notes here...

Submit Form Data

Clear Entries

Document References

Static pages map data out directly without real-time database queries or processor overhead.

Core Benefits:

- Instant browser painting
- Low memory footprints
- High data compatibility

Required Components:

- Document Type Definition
- Semantic Wrapper Blocks
- Data Input Control Interfaces

10. RESULT

Thus, a structured static webpage was successfully designed and developed using HTML5 semantic elements and HTML components. The webpage was verified for its structure, navigation links, form controls, lists, sidebar and footer.